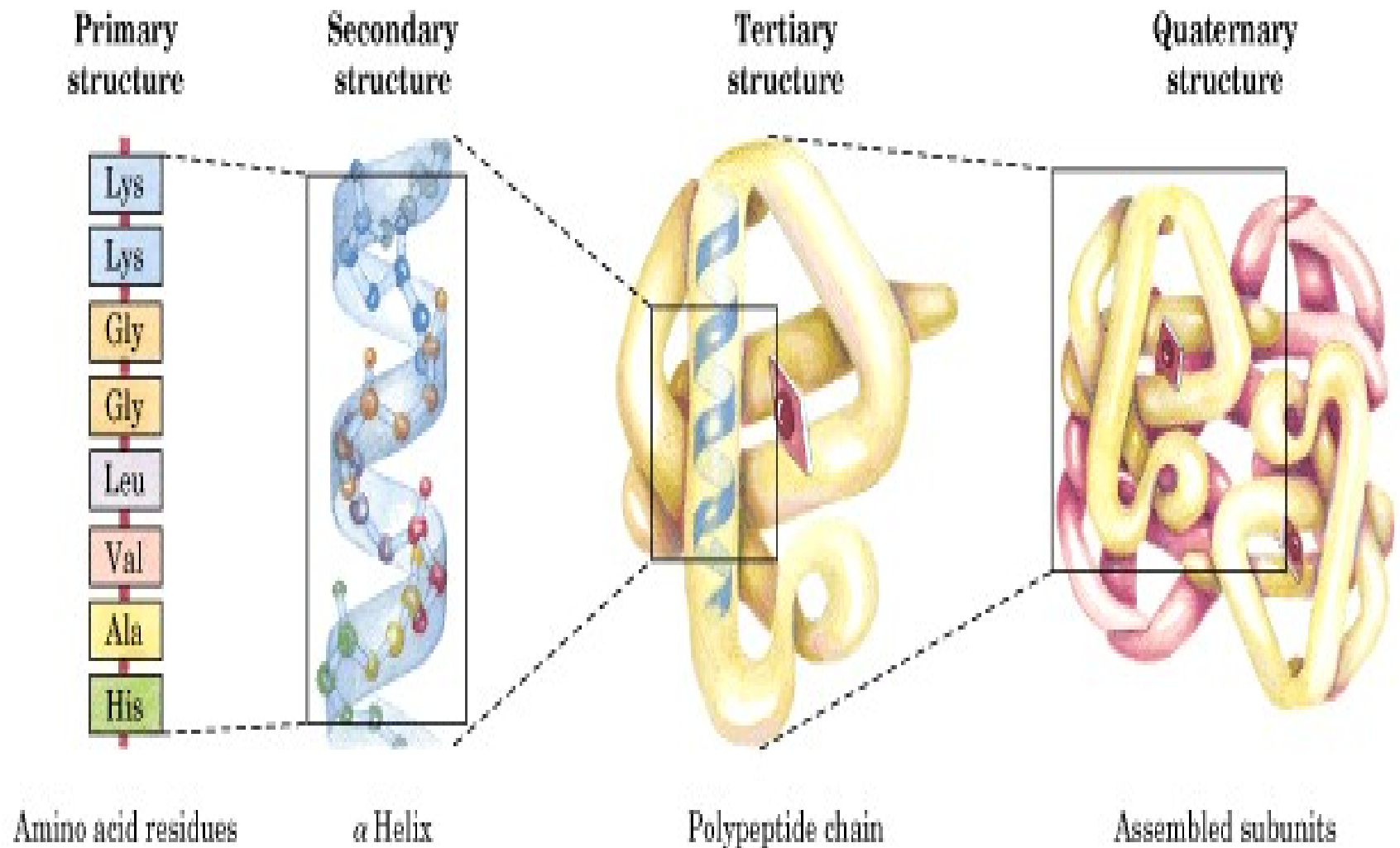


STRUCTURE
MYOGLOBIN & HEMOGLOBIN
OXYGEN & CARBON DIOXIDE TRANSPORT
LUNG SURFACTANT,ROLE OF SURFACTANTS

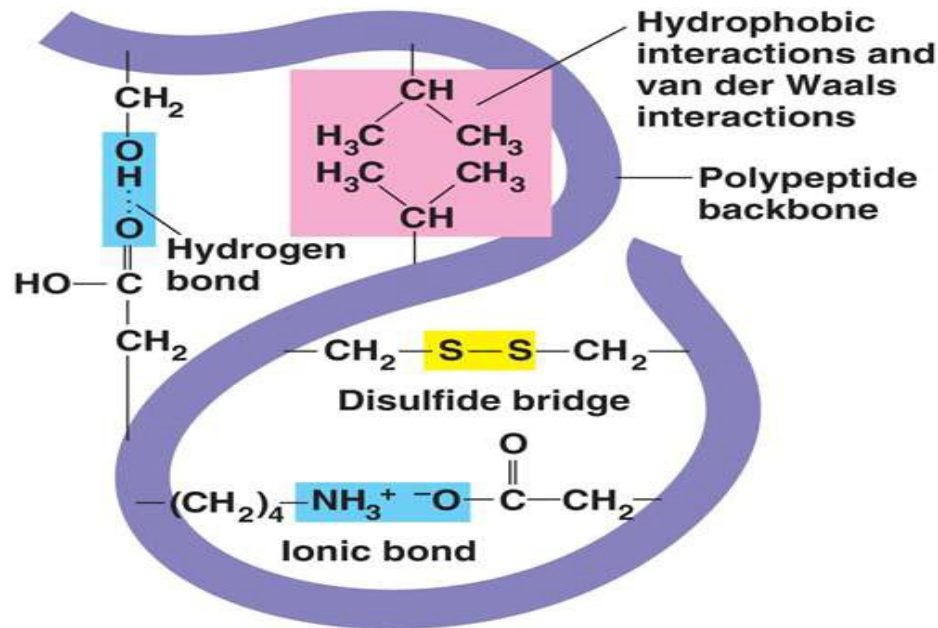
BY : MENGISTU WELDE
(B.pharm, Msc in Bichm)
Asiss.prof

V. On their level of organization



Protein structure

- Majority of functional globular protein have a shape of 3^o structure
- The polypeptide chain is stabilized by a variety of **bonding interactions** of different region of amino acid residues or domain along a single polypeptide chain with their "**side chains of the amino acids** w/c are found in distance



Copyright © 2008 Pearson Education, Inc., publishing as Pearson Benjamin Cummings.

Fig:1 . 3^o structure of polypeptide chain is stabilized by non covalent interaction

Quaternary Structure of protein

Quaternary structure refers an assembly of two or more separate polypeptide chains that are held together

✓ by non-covalent or in some case, covalent interactions

Such protein are know as oligomeric /multimeric

If the subunits /poly peptide chains are identical, it is a homogeneous quaternary structure; but

➤ if there are dissimilarities, it is heterogeneous

Hemoglobin has 4 polypeptide chains,

➤ two of them are α and

➤ two are β , then polymers is heterogeneous tetramers

Humans are aerobic organisms

- Their lungs extract (O_2) from inhaled gases
- Inspired (O_2) leads to a more efficient utilization of fatty acids
- In mammals, Myoglobin is a small monomeric protein, is found primarily in skeletal, striated muscle and
 - mainly serves as a store of O_2 in the cytoplasm &
- from which the organism can draw during increased metabolism or deliver it on demand to the mitochondria

Whereas, Hemoglobin is **restricted** to the erythrocytes

□ which is responsible

➤ for the movement of O_2 from lungs to other tissues

GLOBALAR HEMEPROTEINS

a. Structures of Myoglobin

Hemeproteins are a group of specialized proteins that contain heme as a **tightly bound prosthetic group**

- Its active site is wrapped up by folding of the protein attached to it
- The red color associated with the oxygenated forms of myoglobin and hemoglobin is
 - **due to a heme prosthetic group**
 - **It is essential molecule for the activity of the protein**

Structures of Myoglobin

- Heme = Fe^{2+} (ferrous) + Protoporphyrin
- Protoporphyrin IX = **4 pyrrole rings + 4 methene bridges + 4 methyl groups + 2 vinyl groups + 2 propionate side chains**
- The four pyrrole rings of this system are linked by methene ($-\text{CH}=\text{}$) bridges, so that the unsaturated porphyrin is highly conjugated and planar
- 5th coordination site occupied by the imidazole ring of proximal histidine of myoglobin (or hemoglobin)
- 6th coordination site occupied by oxygen

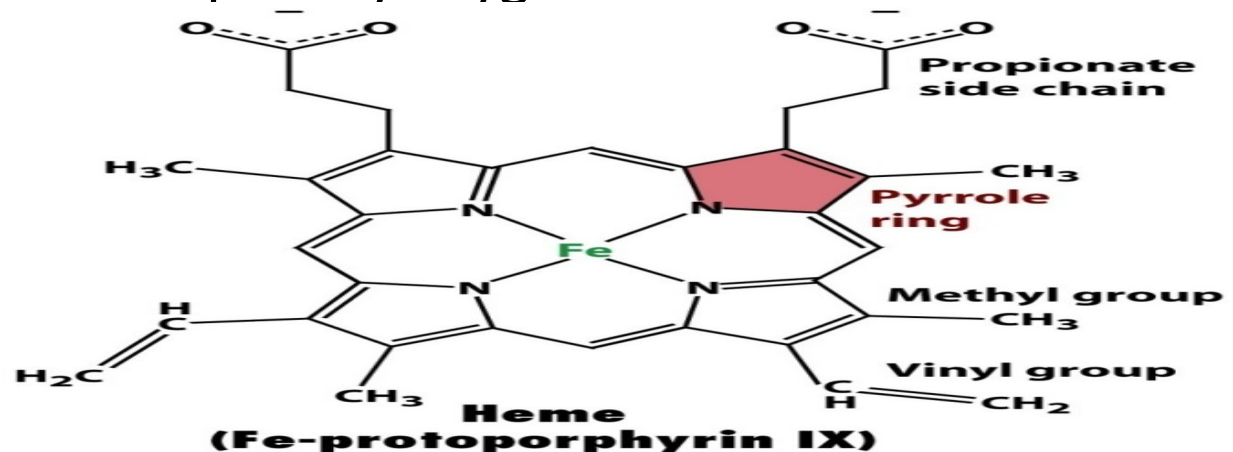


Fig:2 Heme structure

Structures of Myoglobin

- ❖ The bound iron is in the ferrous or Fe(II) oxidation state;
- ❖ it forms a complex with six ligands, four of which are the nitrogen atoms of 4 pyrrole rings of protoporphyrin IX
- ❖ Iron atom coordinated to the side chain of a proximal histidine residue (imidazole on side chain) of the globin molecule, whereas
- ❖ distal histidine residue does not directly interact with the heme group but helps stabilize the binding of oxygen to the ferrous iron or help protect Fe²⁺ oxidation

Structures of Myoglobin

- Accessibility of the heme iron group to molecular oxygen depends on slight movement of these nearby amino acid side chains of globular proteins
- The binding of the porphyrin moiety of the heme to the polypeptide is
 - due to a number of weak interactions, Or none covalent bonds
 - The iron atom of heme is the site of oxygen binding

Structures of Myoglobin

- i. **Myoglobin is a compact molecule, with approximately 80% of its polypeptide chain folded into eight stretches of α -helix**
 - ❖ These α -helical regions, labeled A to H in Figure A

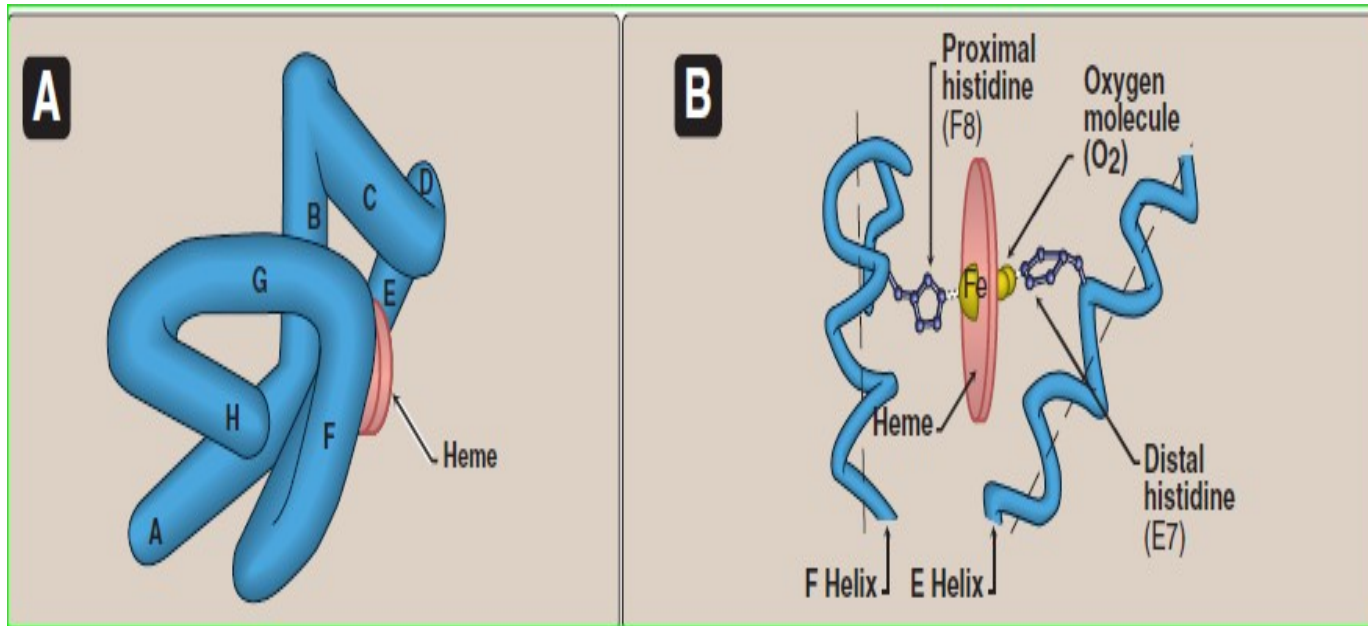


Fig:3 Myoglobin structure

Structures of Myoglobin

- ii. Location of polar and nonpolar amino acid residues:
 - The interior of the myoglobin molecule is,
 - composed almost entirely of nonpolar, highly hydrophobic amino acids
 - E.g. valine, leucine, isoleucine, phenylalanine, and methionine
 - They are packed closely together & stabilized by hydrophobic interactions within the core
 - The surface of the protein contains both hydrophilic and hydrophobic residues e.g. Threonine Tryptophan
 - In contrast;
 - charged amino acids are located almost exclusively on the surface of the molecule, where they can form hydrogen bonds, both with each other and with water.

Structures of Myoglobin & Hemoglobin

- **The protein/ globin**, portion of myoglobin
 - creates a special microenvironment for the heme
- It permits the reversible binding of **one oxygen** molecule (oxygenation) 'But,
- The protein portion of hemoglobin, harbors the heme ring
- It permits the heme molecule to operate efficiently in oxygen binding & transport functions
- The simultaneous loss of electrons by the ferrous iron (oxidation) occurs only rarely

Table 1 Difference between myoglobin & hemoglobin

	HEMOGLOBIN	MYOGLOBIN
Function	O ₂ transport	O ₂ storage
Location	Only in the erythrocyte	Only in skeletal muscle
O ₂ affinity in tissues	Low	High
O ₂ affinity in lungs	High	High
O ₂ affinity change with P _{O₂}	Yes	No
Allosteric regulation	Yes	No
Quaternary structure	Yes—tetramer	No—monomer

b. Structure of Hemoglobin (Hb)

- Hemoglobin is a larger tetrameric metaloprotein that carries oxygen in red blood cells or
- **exclusively** found in erythrocytes (Greek: erythrose—red; kytos—a hollow vessel).
- It is allosteric protein (Fig.1).
- Hemoglobin (mol. wt. 64,450) is a conjugated protein, containing
 - Globin, the apoprotein part and
 - the heme, the non-protein part (prosthetic group)

Structure of Hemoglobin (Hb)

- Hemoglobin A, the major hemoglobin in adults, is composed of four polypeptide, two α & two β chains
- These subunits of hemoglobin are held together by non-covalent interactions primarily hydrophobic, ionic and hydrogen bonds
 - Each subunit of Hb contains a heme group

The normal concentration of Hb in blood

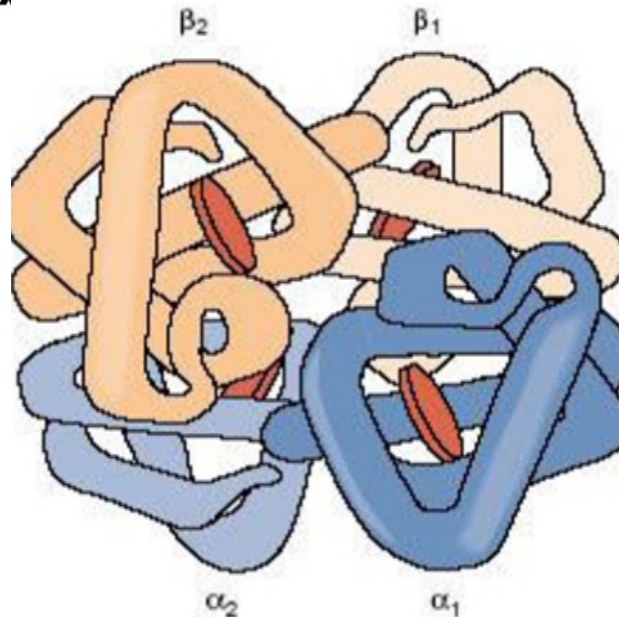
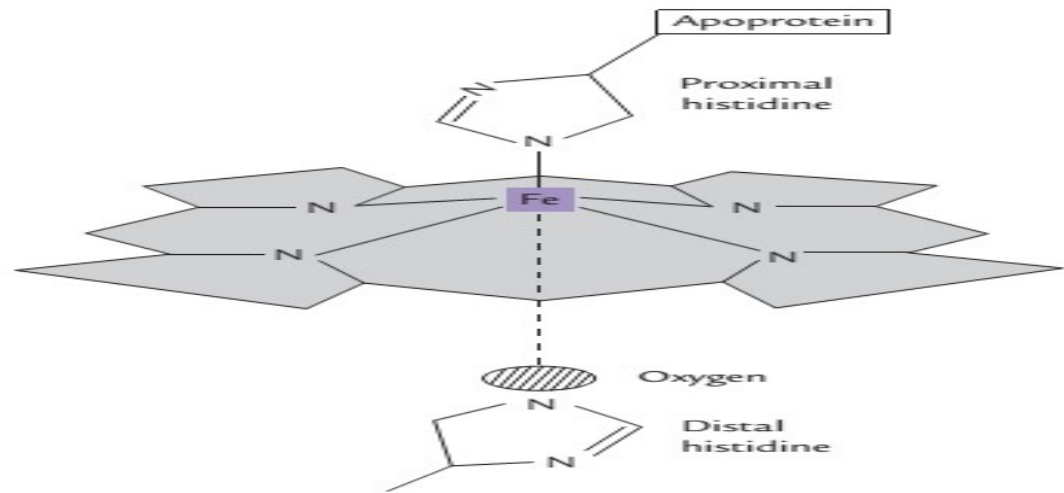
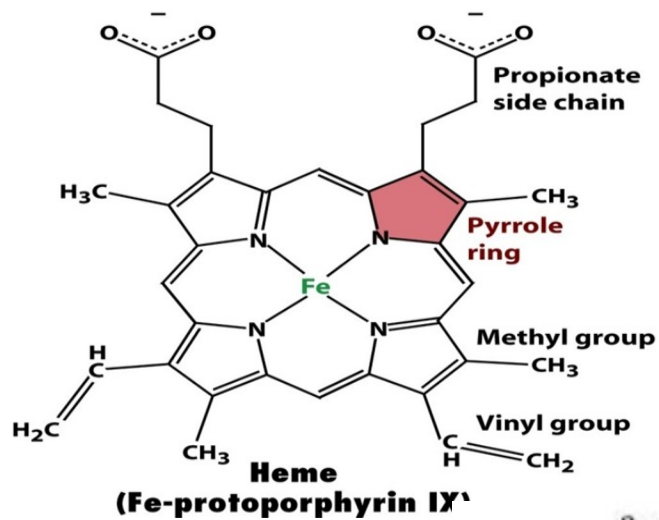
- in males is 14–16 g/dl, and
- in females 13–15 g/dl.

Hemoglobin performs two important biological functions concerned with respiration

1. Delivery of O₂ from the resp. lungs to capillaries of the tissues
2. Transport of CO₂ and protons from tissues to lungs for excretion

Structure of Hemoglobin (Hb) ...

4x



Structure of Hemoglobin (Hb) ...

- Heme is common prosthetic group present in hemoglobin
 - cytochromes, certain enzymes as catalase, tryptophan pyrolase, and chlorophyll (Mg^{2+})
- In case of cytochromes, heme is essential in oxidation and reduction of iron ($\text{Fe}^{2+} \rightarrow \text{Fe}^{3+}$)
 - **is required for e electron transport chain**
- Besides the adult hemoglobin (HbA1) described above, other minor hemoglobins are also found in humans
- **Fetal hemoglobin (HbF)** is synthesized during the fetal development

Structure of Hemoglobin (Hb)

Quaternary structure of hemoglobin:

- Some authors consider hemoglobin consisting of two **identical** dimers
- $(\alpha\beta)_1$ and $(\alpha\beta)_2$, in which **the numbers refer to dimers one and two**

Each α -chain contains 141 amino acids while
 β -chain contains 146 amino acids
of 574



Thus HbA1 has a total

within each dimer the two polypeptide chains are held tightly together, primarily by **hydrophobic interactions**

Note: In this instance, hydrophobic amino acid residues are localized not only in the interior of the molecule, but also in a region on the surface of each subunit

Ionic and hydrogen bonds also occur between the dimers (Figure 5)

Structure of Hemoglobin (Hb)

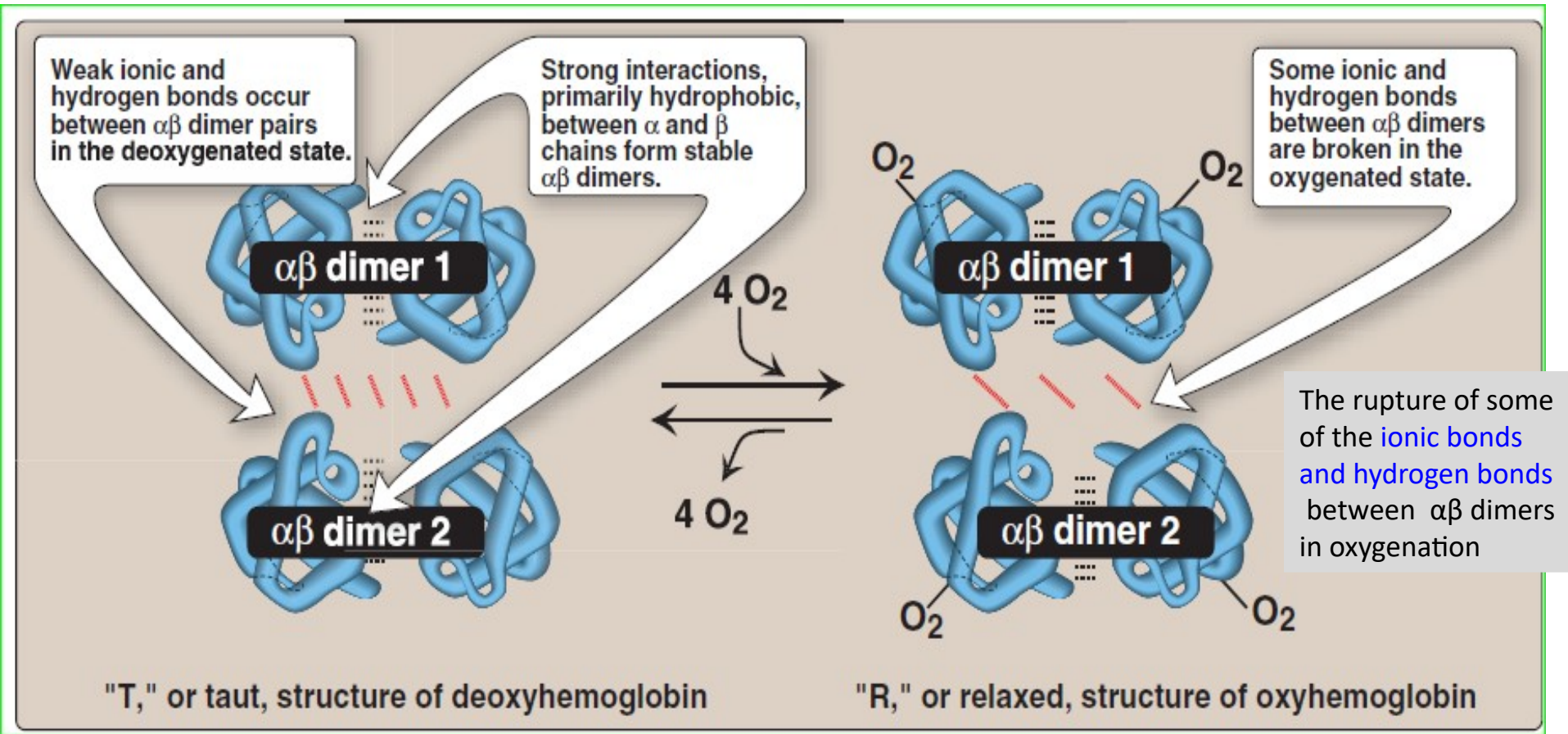


Figure : 5 . Schematic diagram showing structural changes resulting from oxygenation and deoxygenation of hemoglobin

Structure of Hb con't...

Note: The binding of O_2 to the heme iron pulls the iron into the plane of the heme.

- Proximal histidine (F8)

Proximal histidine (F8)

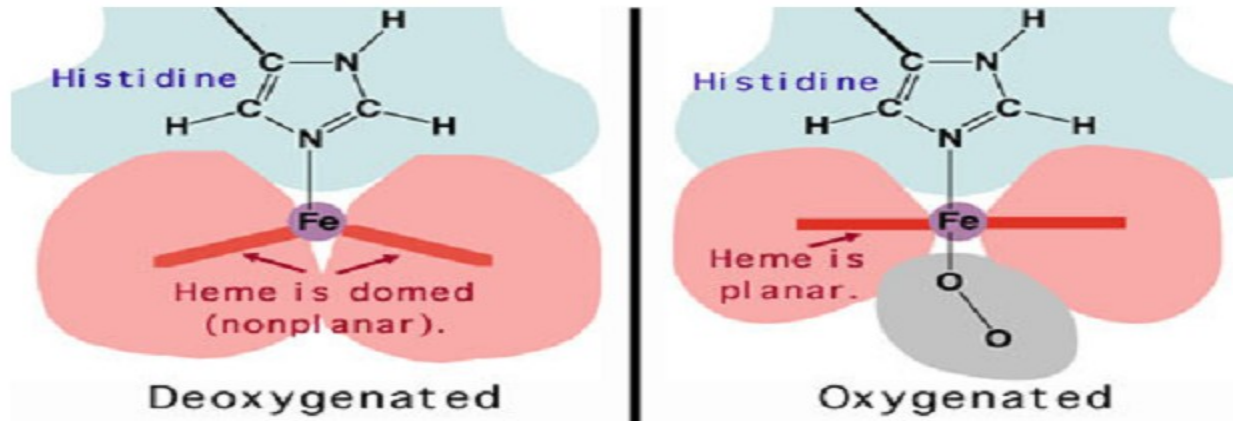


Fig :6. deoxyhemoglobin as compared with oxyhemoglobin

The proximal histidine (F8), also pulls the iron in heme out of the plane of the heme molecule. This iron will be pulled back into the

Oxygen dissociation curve:

Binding of oxygen to myoglobin and hemoglobin

- A plot of Y measured at different partial pressures of oxygen (pO_2) is called the oxygen dissociation curve
- The degree of saturation (Y) of these oxygen-binding sites on all myoglobin or hemoglobin molecules can vary between zero (all sites are empty) and 100% (all sites are full, Fig 7)
- This graph illustrates that myoglobin has a higher oxygen affinity at all pO_2 values than does hemoglobin.

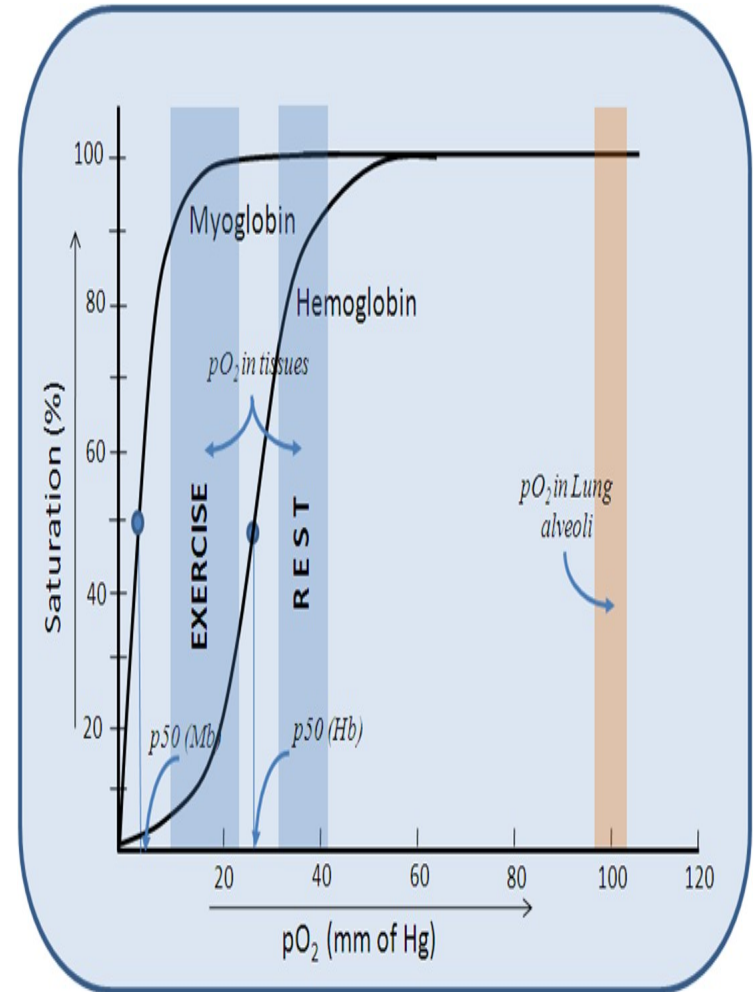
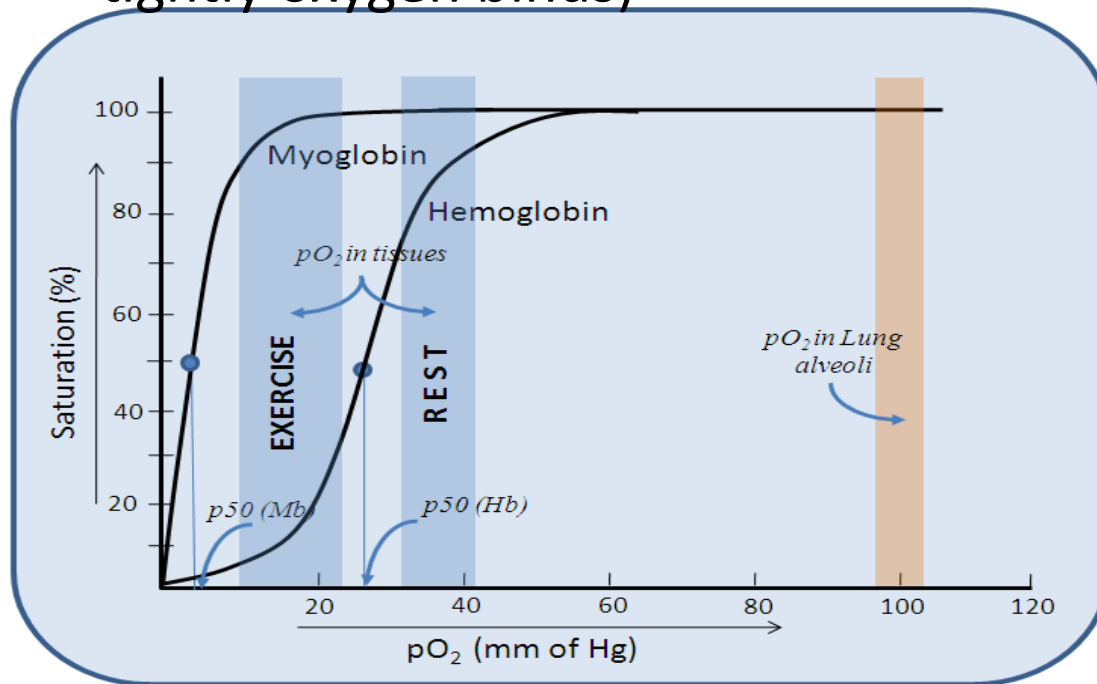


Figure.7.Oxygen dissociation curves for myoglobin and hemoglobin (Hb).

Oxygen binding curves of Mb & Hb

- The partial pressure of oxygen needed to achieve half-saturation (P50) of the oxygen binding sites of myoglobin is approximately 1 mm Hg and 26 mm Hg for hemoglobin.
- The higher the oxygen affinity, the lower the P50 (the more tightly oxygen binds)



The oxygen dissociation curve for hemoglobin is sigmoidal in shape indicating that the subunits cooperate in binding oxygen but myoglobin shows hyperbolic oxygen dissociation curve²²

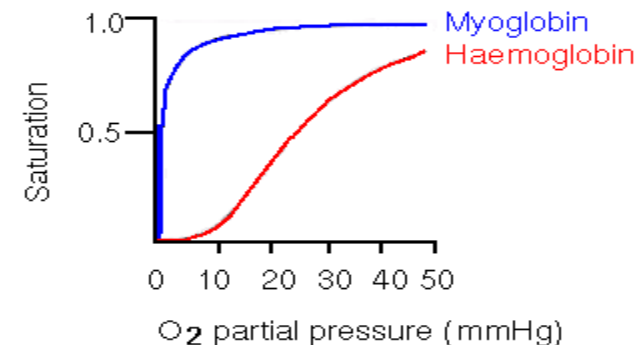


Figure 7. Oxygen dissociation curves for myoglobin and hemoglobin (Hb).

Hemoglobin (Hb):

- The binding of oxygen by hemoglobin is cooperative ; the four protein cannot be independently oxygen-binding subunits
- Cooperative binding of oxygen by the four subunits of hemoglobin means;
 - the binding of an oxygen molecule at one heme group increases the oxygen affinity of the remaining heme groups in the same hemoglobin molecule
- This effect is referred to as heme-heme interaction .
- The affinity for the fourth oxygen to bind hemoglobin is approximately 300 times that of the first.

Although it is more difficult for the first oxygen molecule to bind to hemoglobin, the subsequent binding of oxygen occurs with high affinity, as shown in figure 8.

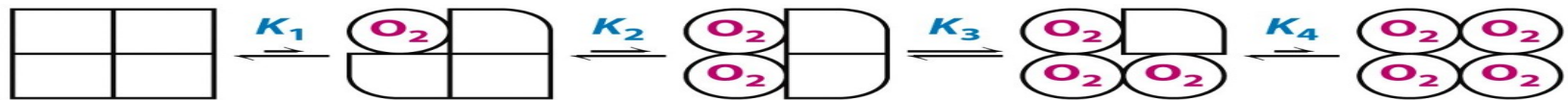


Fig 8. Heme-heme interaction in binding oxygen by the four subunits of hemoglobin

Allosteric effects

i. Hemoglobin is an Allosteric Protein

- The binding and release of oxygen, by hemoglobin are regulated by **allosteric interactions** (from the Greek allos, other).
- In this respect, hemoglobin oxygen carrier protein, **not an enzyme** but resembles certain regulatory enzymes
- The allosteric modulator/effector binds **reversibly** at a site
 - separate from the functional binding site oxygen and
 - **cause significant changes in the conformations of allosteric proteins**

Allosteric effects con't...

- The binding of an allosteric inhibitor
 - to its own allosteric site causes the allosteric protein to change in conformation rapidly from the R state to the T state
- The binding of a substrate to the active site (or an allosteric activator to the allosteric site) causes the reverse effect
- a substrate binds to the functional binding site (the active site)
- Several physiologic factors are responsible or interact with the hemoglobin molecule
 - and facilitate the release of O₂ from oxy-hemoglobin

Clinical Significance

- **They are allosteric effectors of Hemoglobin such as**
 - ❖ 2,3-Bisphosphoglycerate
 - ❖ pH
 - ❖ carbon dioxide and protons, both are products of aerobic metabolism (Bohr Effect)
 - ❖ temperature & Cl⁻
- The curve shift rightward indicates that hemoglobin has a decreased affinity for oxygen, thus, oxygen actively unloads
- A shift of the curve to the left indicates increased hemoglobin affinity for oxygen
e.g. in lungs.

Allosteric effectors

1. 2,3-bis phosphoglycerate (BPG) is an intermediate product of glycolysis

it is produced within the red blood cell that affects hemoglobin's affinity for oxygen.

This short pathway, referred to as Rapaport-Leubering cycle, is described in carbohydrate metabolism

- 2,3-bisphosphoglycerate (2,3BPG) is an allosteric effector
- There is approximately one molecule of BPG per hemoglobin tetramer in the red blood cell
- The concentrations of 2,3BPG and hemoglobin within erythrocytes are nearly equal $[Hb] = [2,3\text{-BPG}] = \sim 2 \text{ mM}$
- 2,3-BPG Specific Binding to T-State Hemoglobin (highly unstable)
- 2,3-BPG can bind to the central pocket located in center of the tetramer
- R to T transition is essential for efficient oxygen unloading

Allosteric effectors ...

- High concentrations of 2,3-BPG will shift the dissociation curve to the right whereas
 - **low concentrations will shift the curve to the left**
- The relationship of hydrogen ions with levels of 2,3 BPG is inversely proportionate
- which stating that an elevation of the hydrogen ion concentration in red blood cells will
 - **result in decreases in 2,3 BPG and vice versa.**
- This is evident at high altitudes where lower oxygen levels induce hyperventilation, causing pCO₂ and hydrogen ions to decrease
 - **which leads to leftward shifting of the dissociation curve.**
- This left shift leads to an increase in red blood cell production of 2,3-BPG,
 - which leads to shifting the curve back to the right and establishes an essential mechanism of respiratory compensation

Allosteric effectors ...

❖ Binding of 2,3BPG to deoxyhemoglobin

The central cavity of deoxyhemoglobin is lined with positively charged groups that are complementary to ;

the carboxylate and phosphate groups of 2,3BPG.

Both help in stabilize the deoxy conformation

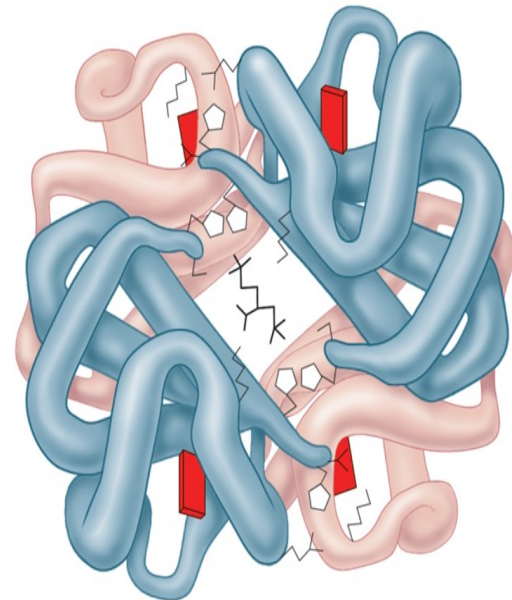
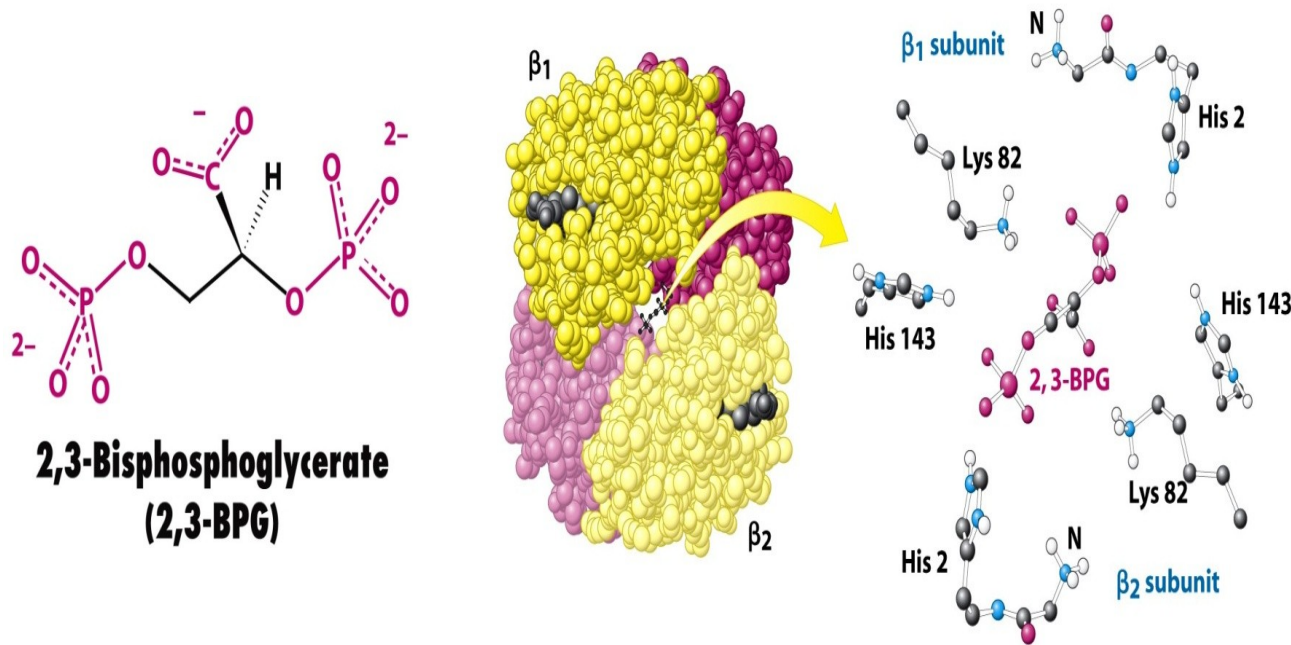


Figure :8. The α subunits are shown in pink, the β - sub units in blue, and the heme prosthetic groups in red.

Mechanism of action of 2,3-BPG

- One molecule of 2,3-BPG binds with one molecule (tetramer) of deoxyhemoglobin (which has a low affinity) in the central cavity of the four subunits
- Ionic bonds (salt bridges) are formed between;
 - the positively charged amino acids (of β -globins) with
 - the negatively charged phosphate groups of 2,3-BPG by crosslinking the β - chains. (Fig.8).
- In the central cavity of hemoglobin, between the **two β subunits**, there are **six positively charged side chains** and the N-terminal α -group of each β chain, can interact electrostatically with the **five negative charges** of 2,3BPG.

Allosteric effectors con't...

- In oxyhemoglobin, the chains are closer together, and the allosteric binding site is too small to accommodate 2,3BPG
- **Oxygen binding** increases the proportion of hemoglobin molecules in the **oxy (R) conformation**, and
- **2,3BPG binding** increases the proportion of hemoglobin molecules in the **deoxy (T) conformation**.
- Because oxygen and 2,3BPG have different binding sites, 2,3BPG **is an allosteric effector**

NB : 1g Hb binds with 1.34 ml O₂, so 15g Hb x 1.34 ml O₂ = 20.1 ml of O₂/100 ml blood

Allosteric effectors ...

2,3-BPG regulates the binding of O₂ to hemoglobin

It specifically binds to deoxyhemoglobin (and not to oxyhemoglobin) and decreases the O₂ affinity to Hb

- The effect of 2,3-BPG on Hb may be summarized as follows:
 - ✓ $\text{HbO}_2 \text{ (OxyHb)} + 2,3\text{-BPG} \rightarrow \text{Hb-2,3-BPG (DeoxyHb bound to 2,3-BPG)} + \text{O}_2$
- It reduced affinity of O₂ to Hb facilitates the release of O₂ in the tissues
- This 2,3-BPG shifts the oxygen dissociation curve to the right

Clinical significance of 2,3-BPG

In erythrocyte, the levels of 2,3-BPG are related to tissue demands of oxygen supply

high altitude adaption

As we go up high altitude or high mountain the concentrations of oxygen is more and more decrease .

The situation rise 2,3BPG synthesis which facilitate the release of oxygen to tissue

2. **Air way pulmonary disease:** during this diseases the air way is narrow & oxygen provided to hemoglobin is less ,so under such situation RBC must be increase in 2,3BPG synthesis , in

order to crosslink the two beta chain of hemoglobin to better delivery of oxygen to tissue

4. **Sever anemia:** concentration of hemoglobin is low to provide enough oxygen to tissue, so the amount of 2,3BPG synthesis by RBC must be increase in order to crosslink the **two β -chain** of hemoglobin & better delivery of oxygen to tissue

Clinical significance of 2,3-BPG

4. Blood stored for transfusion:

Stored blood Lose of 2,3 BPG & needs to be reconstituted with Inosine w/c contain **hypoxantin ribose** , Addition of inosine (hypoxanthine-ribose) to the stored blood prevents the decrease of 2,3-BPG so blood transfused to victims provide O₂ to tissue

Allosteric effectors con't...

2. Bohr Effect : protons and Carbon dioxide

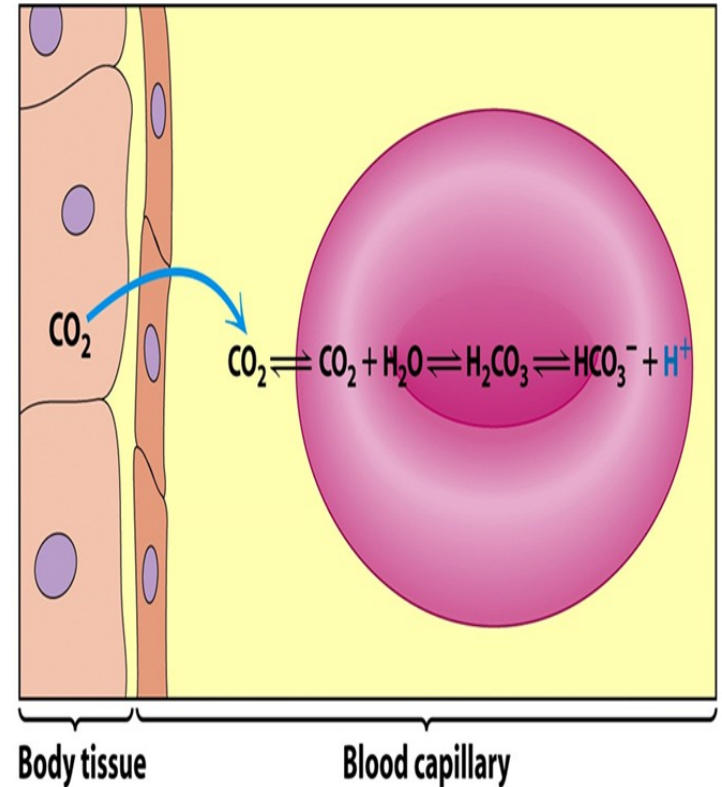
Regulation of the binding of oxygen to hemoglobin by **carbon dioxide and protons**

The binding of oxygen to hemoglobin decreases with increasing H^+ concentration (lower pH) or when the hemoglobin is exposed to increased partial pressure of CO_2 (pCO_2)

- This phenomenon is known as Bohr effect
- Bohr effect causes oxygen dissociation curve to a shift to the right

In Tissue

- The majority of the CO₂ released from the peripheries tissues enters the blood , the enzyme carbonic anhydrase present in erythrocytes catalyzes the formation of carbonic acid (H₂CO₃),w.c later dissociates
- CO₂ decreases the affinity of hemoglobin for O₂ in peripheral tissues by lowering the pH inside red blood cells

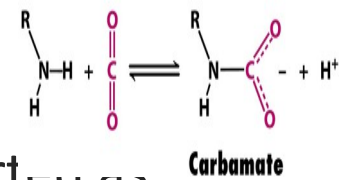


This is in metabolically active tissue metabolizes glucose by oxygen into CO₂ and organic acids

The Bohr Effect : Carbon Dioxide con't...

Carbon Dioxide

- In aerobic metabolism, for every molecule of O₂ utilized by tissues , one molecule of CO₂ is liberated
- The majority of Carbon dioxide is transported from the tissues to the lungs in two ways.
 - ✓ About 10% as dissolved form in plasma
 - ✓ About 15% of CO₂ carried in blood directly binds with Hb in the form of **CARBAMINOHEMOGLOBIN** (carbamate adducts)
 - ✓ It is none covalent bound to globulin chain of hemoglobin & transport carbon dioxide in blood
 - ✓ $\text{Hb} - \text{NH}_2 + \text{CO}_2 = \text{Hb} - \text{NH} - \text{COO}^- + \text{H}^+$
 - ✓ About 75% of CO₂ of the tissue, or most CO₂ is transported as dissolved bicarbonate ions (HCO₃⁻)



The Bohr Effect : Carbon Dioxide

Hemoglobin acts as a buffer and immediately binds with protons

It is estimated that for every 2 protons bound to Hb, 4 oxygen molecules are released to the tissues

In the lungs

- binding O₂ to Hb results in the release of CO₂ and H⁺ from Hb
- reaction reverses Low pCO₂;
- The bicarbonate and protons combine to form carbonic acid, to release CO₂, which is exhaled

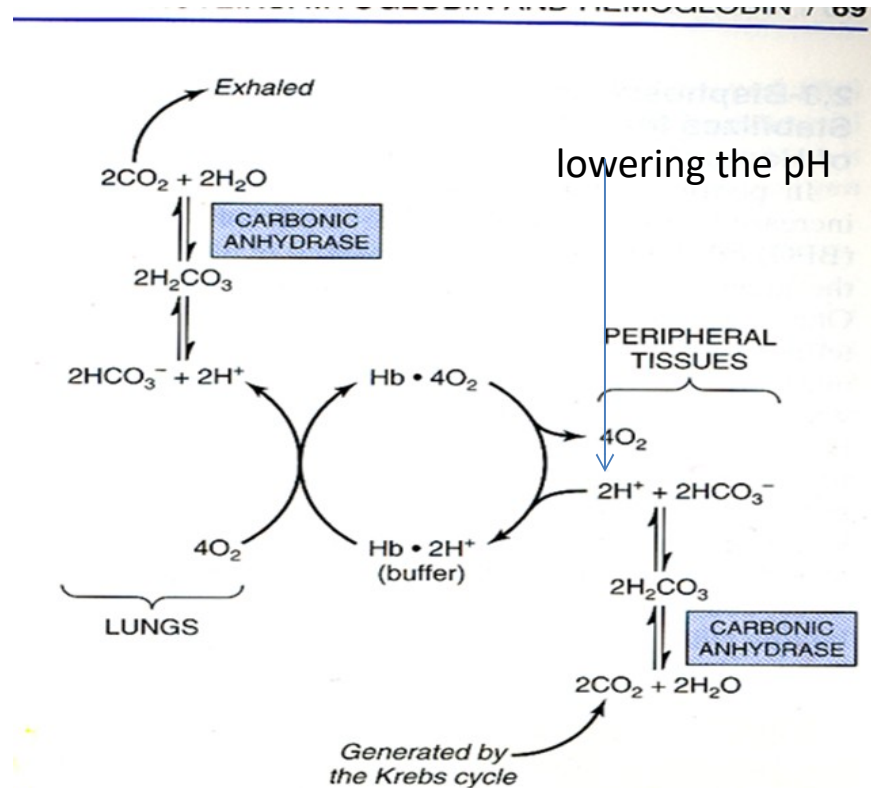
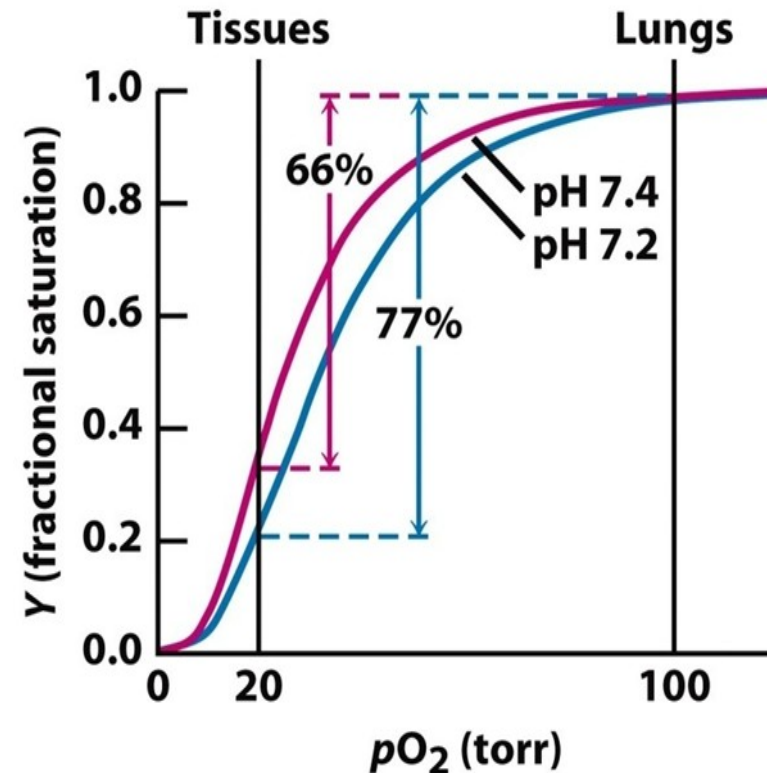


Figure 7-15. The Bohr effect. The carbon dioxide generated in peripheral tissues combines with water to form carbonic acid, which dissociates into protons and bicarbonate ions. The deoxygenated hemoglobin acts as a buffer by binding protons and delivering them to the lungs. In the lungs, the binding of oxygen by hemoglobin releases protons that combine with bicarbonate ion, forming carbonic acid, which when dehydrated by carbonic anhydrase becomes carbon dioxide, which is then exhaled from the lungs.

Allosteric effectors con't...

- The increase in the concentration of CO₂ and the concomitant decrease in pH, raise the P₅₀ of hemoglobin.
- This phenomenon, called the Bohr effect, increases the efficiency of the oxygen delivery system.

while an increase in pH (alkalinity) shifts the dissociation curve to the left.
NB. In inhaling lungs, where the CO₂ level is low, O₂ is readily picked up by hemoglobin



Decrease of pH from 7.4 to 7.2 increases oxygen unloading efficiency by 11%.

Temperature:

- The effect of temperature on the curve
- Higher temperatures favored unloading O_2 from Hb & cause a rightward shift.
- On the other hand, lower temperatures will cause a leftward shift in the dissociation curve
- A notable example of this is exercise, where the temperature of muscle increases secondary to its utilization, thus
 - shifting the curve to the right and allowing oxygen to be more easily unloaded from hemoglobin and deliver it to tissues in need.

(Cont...) Blood, Hb and O₂-saturation

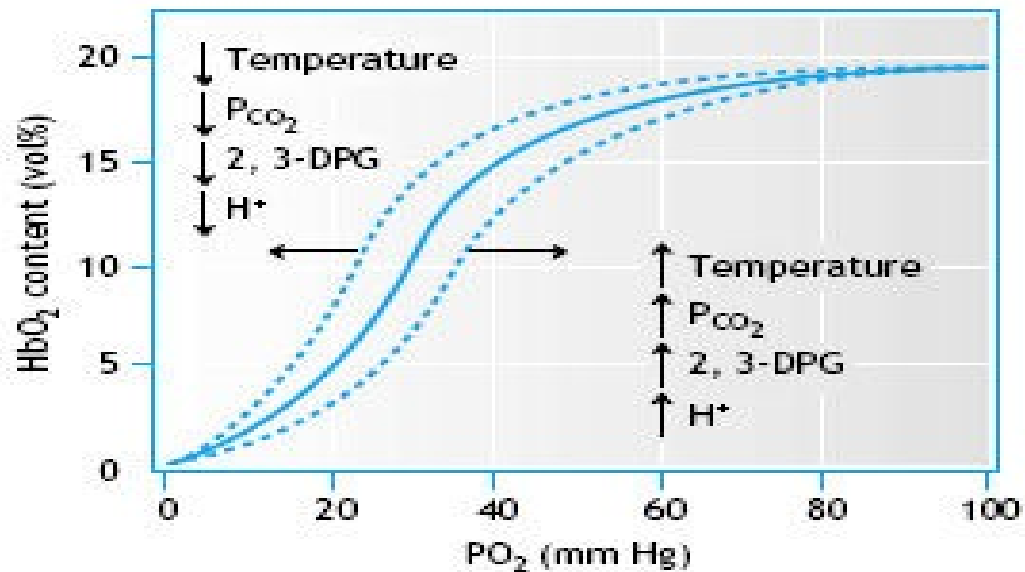


Figure 3–7. Changes in affinity of hemoglobin (Hb) for O₂ (oxyhemoglobin dissociation curve).

NB. pH, Temp, 2,3-BPG affects Hb saturation with O₂.

- Hemoglobin (specifically heme) combines with different ligands and forms hemoglobin derivatives.
- The normal blood contains oxyHb and deoxyHb.
 - Besides these, methemoglobin (metHb) and carboxyhemoglobin are the other important Hb derivatives.

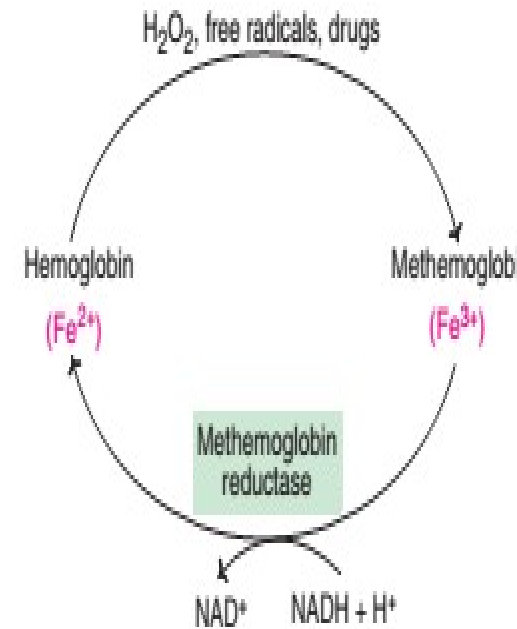
Hemoglobin derivatives

Carboxyhemoglobin (COHb) :

- Carbon monoxide (CO) is a toxic compound (an industrial pollutant) that can bind with Hb in the same manner as O₂ binds. due to the competition of the same binding sites.
- However, CO has about 200 times more affinity than O₂ for binding with Hb
- Clinical manifestations of CO toxicity are observed when the COHb concentration exceeds 20%
- Binding of one CO molecule to hemoglobin increases the affinity of the other binding spots for oxygen, leading to a left shift in the dissociation curve.

Methemoglobin

- For the biological function of hemoglobin, to carry oxygen
 - **the iron should remain in the ferrous (Fe^{2+}) state.**
- Hemoglobin (Fe^{2+}) can be oxidized to methemoglobin (Fe^{3+}).
- In normal circumstances, however, molecular oxygen does not oxidize Hb, it only loosely binds to form oxyhemoglobin.
- The oxidation of hemoglobin to methemoglobin (metHb) may be caused in the living system by H_2O_2 , free radicals and drugs
- The methemoglobin (with Fe^{3+}) is unable to bind to O_2 ,
 - **Instead, a water molecule occupies the oxygen site in the heme of metHb.**



Hemoglobin derivatives...

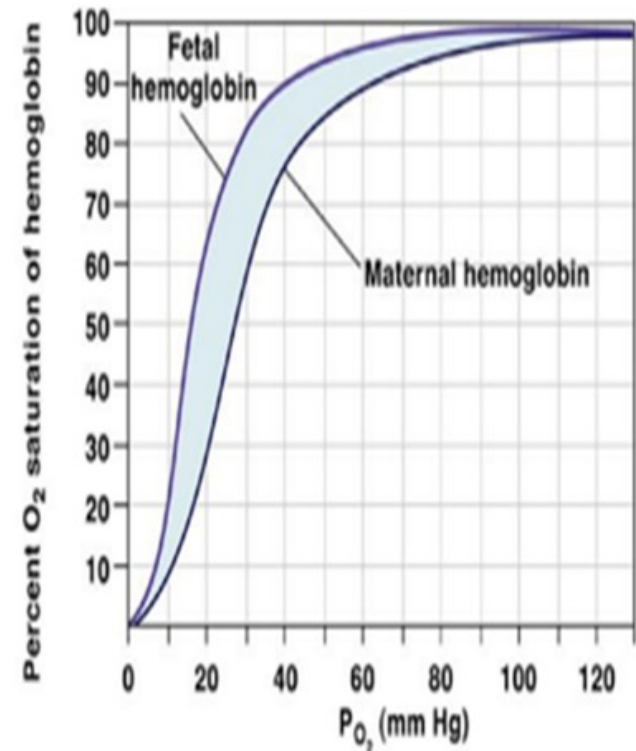
Glycosylated hemoglobin (HbA1c)

- Is produced when glucose reacts with the amino group on a hemoglobin molecule forming a ketoamine.
- The glucose molecule is attached to one or both N-terminal valines of the β -chains of normal adult hemoglobin
- It is increased in diabetes mellitus which is successfully utilized for the prognosis/ predicting of these patients

Allosteric effectors con't...

Fetal hemoglobin (HbF) :

- Fetal hemoglobin (HbF) is structurally different from adult hemoglobin because it is composed of two alpha (α) and two gamma chains (γ)
- The gamma chains of HbF have a reduced affinity for 2,3-BPG, thus allowing HbF to have a higher affinity for oxygen at lower levels of P_{O_2} and resulting in a leftward shift of the dissociation curve.
- This may be needed for the transfer of oxygen from the maternal blood to the fetus



- Hemoglobinopathies result from **abnormalities in the hemoglobin molecule**
- genetic mutations or disorders passed down through families in which there is;
 - abnormal synthesis or abnormal structure of the hemoglobin molecule or rarely both
- Abnormalities may due to changes in the normal amino acid sequence of globin
- eg. Sick cell anemia, thalassemia, hemoglobin C disease
- Sick cell anemia (HbS) : is the most common form of abnormal hemoglobins in w/c, glutamic acid is replaced by valine at Beta-chain and
- Sick cell disease is a group of disorders that affects hemoglobin, the molecule in red blood cells that delivers oxygen to cells throughout the body.



Hemoglobinopathies

- Hemoglobin C is a hemoglobinopathy that is hereditary
- Hemoglobin C disease (HbC) are the classical examples of abnormal hemoglobins
- It is caused due to mutation in the beta-globin chain in which glutamate (acidic) is replaced by lysine (basic) in the sixth position of the beta-globin chain.
- This mutation makes Hb C less soluble than Hb A, forming hexagonal crystals
- a condition affecting a protein in the blood (hemoglobin) which transports oxygen throughout the body.

Hemoglobinopathies ...

Thalassemias

- are a heterogeneous grouping of genetic disorders /mutation that result from a decreased synthesis of alpha or beta chains of hemoglobin
- Hemoglobin serves as the oxygen-carrying component of the red blood cells.
- It consists of two proteins, an alpha, and a beta. If the body does not manufacture enough , one or the other of these two proteins, the red blood cells cannot carry sufficient oxygen to tissue ;
 - this causes anemia that begins in early childhood and lasts throughout life.
- Thalassemia is an inherited disease, meaning that at least one of the parents must be a carrier for the disease, which leads to severe anemia

Classified according to globin chain that is affected

- Alpha thalassemia is caused by reduced or absent synthesis of alpha globin chains, of the hemoglobin and
- beta thalassemia is caused by reduced or absent synthesis of beta globin chains
- Imbalances of globin chains cause hemolysis and impair hemoglobin In erythropoiesis

Lung surfactant

- Pulmonary surfactant is a complex mixture of phospholipids and proteins which is synthesized and secreted by alveolar type II cells in the airways
 - It **consists of 90% phospholipids and 10% surfactant proteins**

Composition

- ~40% dipalmitoylphosphatidylcholine (DPPC)
- ~40% other phospholipids (PC)
- ~10% neutral lipids (Cholesterol)
- ~10% surfactant proteins (SP-A, SP-B, SP-C and SP-D)
- The main phospholipid in surfactant is dipalmitoylphosphatidylcholine (DPPC), also known as lecithin
 - DPPC is a phospholipid with two 16-carbon saturated chains and a phosphate group with quaternary amine group attached



- DPPC also has a higher compaction capacity than the other phospholipids,
 - because the a polar tail is less bent.
- the DPPC's adsorption kinetics is very slow
- the surfactant have both hydrophilic and hydrophobic /amphipathic properties
- Which help lung surfactant allow its alignment at the air-liquid interface.
- By adsorbing to the air-water interface of alveoli,
 - with hydrophilic head groups in the water and
 - the hydrophobic tails facing towards the air,

Lung surfactant...

It acts to decrease surface tension at the air–liquid interface of the alveoli

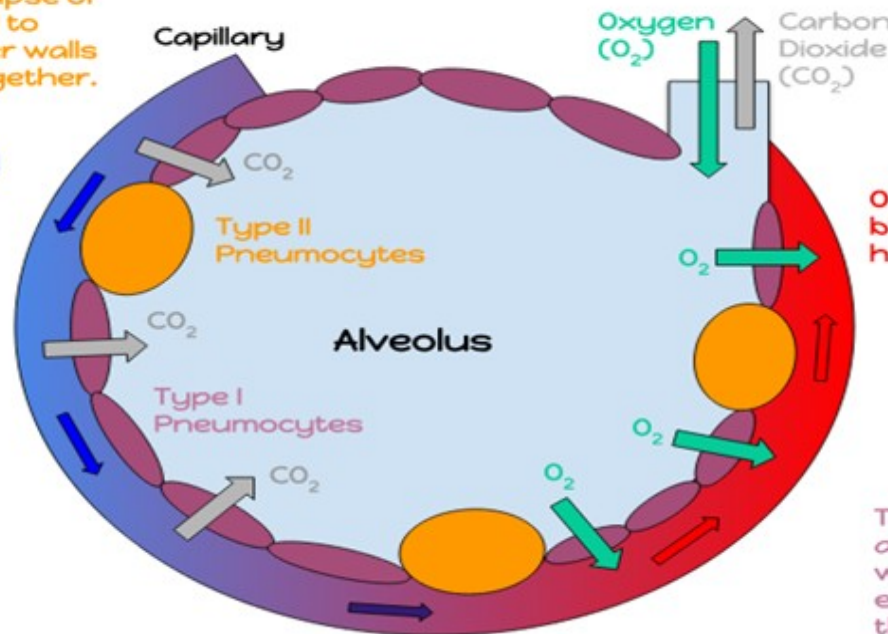
It begins to be produced in the fetus at about 24 to 28 weeks of pregnancy, and is found in;

- ✓ amniotic fluid between 28 and 32 weeks

- The alveolar membrane is the gas exchange surface, surrounded by a network of capillaries
- Oxygen is diffused across the membrane into the capillaries and carbon dioxide is released from the capillaries into the alveoli to be breathed out

Type II pneumocytes secrete surfactant to prevent the collapse of the alveolus and to prevent the inner walls from sticking together.

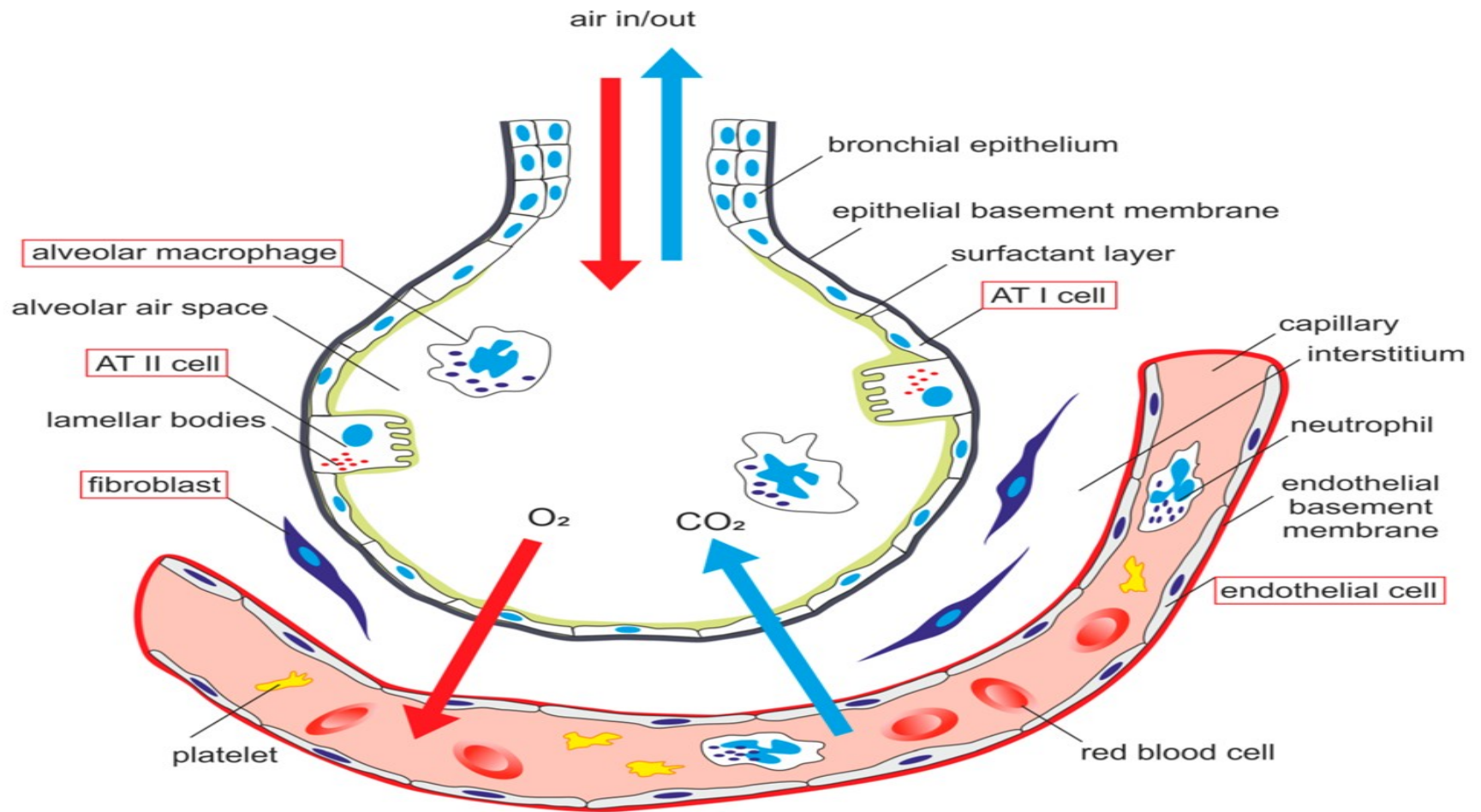
Deoxygenated blood coming from heart



Oxygenated blood going to heart

Type I Pneumocytes are thin, flat cells which allow gas exchange between the alveolus and capillaries.

Fig An annotated diagram of the alveolus



Lung surfactant...

- The number of type II cells increases until the 32 week,
- The time at which surface active agent appears in, the lung and amniotic fluid.
- Surface tension decreases when inclusion bodies increase in the type II cells or surfactant producing pneumocytes
- In the few weeks before term, one can perform screening tests on amniotic fluid to detect,
- newborns that are at risk for respiratory distress syndrome (RDS)
- For predicting pulmonary lung maturity, the critical L/S ratio, is 2.0 or greater
- If the tests is positive for RDS, the mother should be treated with a glucocorticoid drug to accelerate maturation of the fetal lung.

Role of Lung surfactant

- Surfactant lowers alveolar surface tension and is crucial for the prevention of alveolar collapse, or
- To prevent atelectasis (collapse of the alveoli or atriums) at the end of expiration.
- In addition,
 - surfactant contributes to airway muco-ciliary clearance.
 - reduces fluid accumulation and keeps the airways dry by reducing surface tension
- Surface tension draws **fluid from capillaries** to the alveolar spaces
- Surfactant-specific proteins are part of the innate immune defense mechanisms of the lung.

Proteins

- Proteins make up the remaining 10% of the surfactant.
- Half of this 10% is plasma proteins but the rest is formed
- By the apolipoproteins, surfactant proteins SP-A, SP-B, SP-C, and SP-D.
- The apolipoproteins are produced by the secretory pathway in type II cells.

SP-A and SP-D are collectins or binding to microorganisms

They confer innate immunity as they have carbohydrate recognition domains that allow them to coat bacteria and viruses promoting phagocytosis by macrophages & defense the lung

SP-B and SP-C are hydrophobic membrane proteins that increase the rate that surfactant spreads over the surface.

SP-B and SP-C are required for the proper biophysical function of the lung

- A pulmonary alveolus (plural: alveoli, from Latin alveolus, "little cavity"), also known as an air sac or air space, or cavities in the lungs where gas exchange takes place.
- Oxygen is exchanged for carbon dioxide at the blood–air barrier between the alveolar air and the pulmonary capillary
- They are located sparsely in these bronchioles, line the walls of the alveolar ducts, & are more numerous in blind-ended alveolar sacs

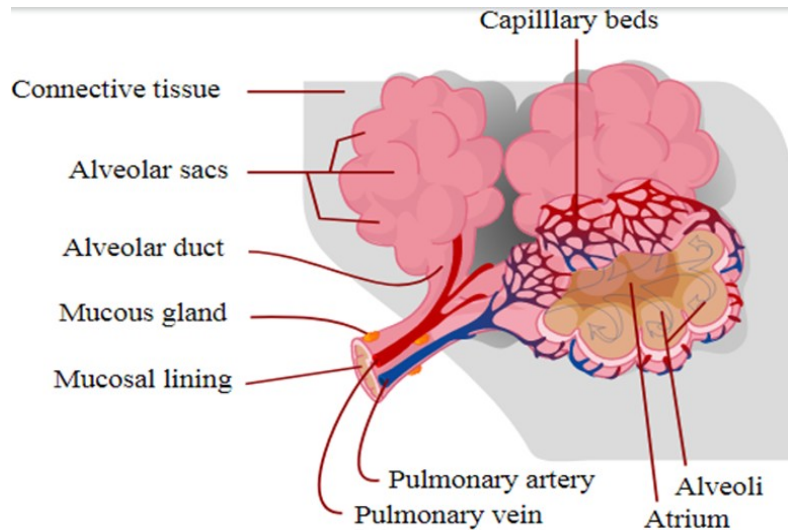
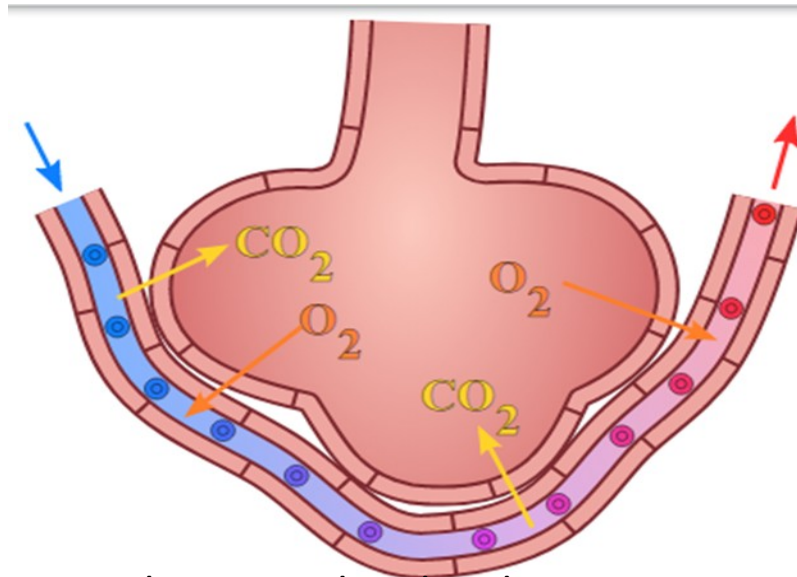
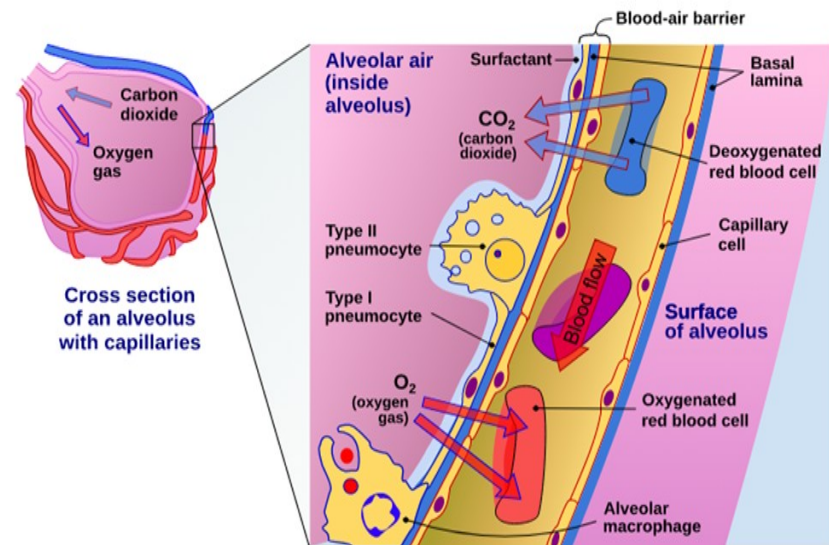


Fig An anatomical structure alveolus (plural: alveoli, from Latin alveus, "little a hollow cavity") Mainly found in the lung, the pulmonary alveoli are spherical of the respiratory bronchioles and are the primary sites of gas exchange with the blood.



Gas exchange in the alveolus



The cross section of an alveolus with capillaries is shown. Part of the cross section is magnified to show diffusion of oxygen gas and carbon dioxide through type I cells and capillary cells.

Summary of Clinical significance Surfactant

- Insufficient surfactant in the alveoli is one of the causes that can contribute to atelectasis (collapse of part or all of the lung).
- The severe condition of acute respiratory distress syndrome (ARDS) is caused by a deficiency or dysfunction of surfactant
- Insufficient surfactant in the lungs of preterm infants causes infant respiratory distress syndrome (IRDS)
- The lecithin–sphingomyelin ratio is,
 - a measure of fetal amniotic fluid to indicate lung maturity or immaturity
- A low ratio indicates a risk factor for IRDS.
- Lecithin and sphingomyelin are two of the phospholipid of pulmonary surfactant.

Clinical significance ...

- Lung surfactant alterations have been described in a number of respiratory diseases.
- Surfactant deficiency in premature animals causes neonatal respiratory distress syndrome

Mechanisms of Lung Surfactant Abnormalities

- Surfactant deficiency can be due to :
 - an immaturity or injury to type II alveolar cells leading to insufficient surfactant synthesis, storage, or release

Clinical significance ...

- Surfactant dysfunction or loss of surface activity can be caused by the degradation of surfactant components
- Degradation of surfactant components occurs via the
 - action of proteases, phospholipases, or reactive oxygen species liberated during pulmonary inflammation
- [Read more on Hemoglopathy & Lung surfactant](#)

Thank you !!